

JS Objects Literals

Used to store keyed collections & complex entities.

property => (key, value) pair

Objects are a collection of properties

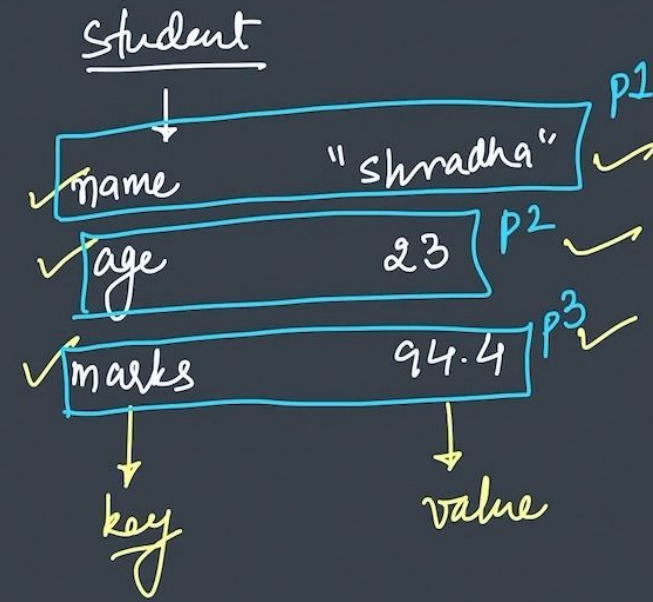


JS Objects Literals

Used to store keyed collections & complex entities.

property => (key, value) pair

Objects are a collection of properties



JS Objects Literals

```
let delhi = {  
  latitude: "28.7041° N",  
  longitude: "77.1025° E"  
};
```

```
const student = {  
  name: "shradha",  
  age: 23,  
  marks: 94.4,  
  city: "Delhi"  
};
```



<> index.html

JS app.js



JS app.js > ...

```
1  const student = {  
2      name: "shradha",  
3      age: 23,  
4      marks: 94.4  
5  };
```



<> index.html

JS app.js



JS app.js > [🔗] item > 🔑 discount

```
1  const student = {
2    name: "shradha",
3    age: 23,
4    marks: 94.4
5  };
6
7  const item = {
8    price: 100.99,
9    discount: 50,
10   colors = ["red", "pink"]
11  };
```

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Thread / Twitter Post

Create an object literal for the properties of thread/ twitter post which includes -

- username
- content
- likes
- reposts
- tags



<> index.html

JS app.js



JS app.js > [🔗] post > 🔑 tags

```
1  const post = {  
2    username: "@shradhakhapra",  
3    content: "This is my #firstPost",  
4    likes: 150,  
5    reposts: 5,  
6    tags: ["@apnacollege", "@delta"]  
7  };
```



JS

File

/Users/shradhakhapra/WebClassroom/JavaScript/index.html

Elements

Console

Sources

Network

Performance

top

Filter

Default levels

No Issues

> post

< {username: '@shradhakhapra', content: 'This is my #firstPost', likes: 150, reposts: 5, tags: Array(2)}

> |

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APNA COLLEGE



JS x +

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Elements Console Sources Network Performance >>

top Filter Default levels No Issues

```
> post
< {username: '@shradhakhapra', content: 'This is my #firstPost', likes: 150, reposts: 5, tags: Array(2)}
> post["content"];
< 'This is my #firstPost'
> |
```



```
Elements Console Sources Network Performance >>
top Filter Default levels No Issues
> post
< {username: '@shradhakhapra', content: 'This is my #firstPost', likes: 150, reposts: 5, tags: Array(2)}
> let prop = "repost";
< undefined
> post[prop];
< undefined
> let prop = "reposts";
< undefined
> post[prop];
< 5
> post.prop;
< undefined
>
```

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Get Values

JS automatically converts objects keys to strings.

Even if we made the number as a key, the number will be converted to string.



JS x +

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Elements Console Sources Network Performance >>

top Filter Default levels No Issues

```
> obj
< {1: 'a', 2: 'b', true: 'c', null: 'd', undefined: 'e'}
  1: "a"
  2: "b"
  null: "d"
  true: "c"
  undefined: "e"
  ▶ [[Prototype]]: Object
> "null"
< 'null'
> null
< null
>
```



JS x +

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Elements Console Sources Network Performance >>

top Filter Default levels No Issues

```
> obj
< {1: 'a', 2: 'b', true: 'c', null: 'd', undefined: 'e'}
  1: "a"
  2: "b"
  null: "d"
  true: "c"
  undefined: "e"
  ▶ [[Prototype]]: Object

> "null"
< 'null'

> null
< null

> obj[1]
< 'a'

> obj[2]
< 'b'

> obj[null]
< 'd'

> obj[true]
< 'c'

> obj[undefined]
< 'e'

>
```

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Add/ Update Value

- Change the city to "Mumbai"
- Add a new property, gender: "Female"
- Change the marks to "A"

```
const student = {  
  name: "shradha",  
  age: 23,  
  marks: 94.4,  
  city: "Delhi"  
};
```



JS x +

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Elements Console Sources Network Performance >> | Settings | Close

top Filter Default levels No Issues

```
> student
< ▶ {name: 'shradha', age: 23, marks: 94.4, city: 'Delhi'}

> student.city
< 'Delhi'

> student.city = "Mumbai";
< 'Mumbai'

> student.city
< 'Mumbai'

> student
< ▶ {name: 'shradha', age: 23, marks: 94.4, city: 'Mumbai'}

> student.gender
< undefined

> student.gender = "female";
< 'female'

> student.gender
< 'female'

> student
< ▶ {name: 'shradha', age: 23, marks: 94.4, city: 'Mumbai', gender: 'female'}
```



```
Elements Console Sources Network Performance >>
top Filter Default levels No Issues
> student.city
< 'Delhi'
> student.city = "Mumbai";
< 'Mumbai'
> student.city
< 'Mumbai'
> student
< {name: 'shradha', age: 23, marks: 94.4, city: 'Mumbai'}
> student.gender
< undefined
> student.gender = "female";
< 'female'
> student.gender
< 'female'
> student
< {name: 'shradha', age: 23, marks: 94.4, city: 'Mumbai', gender: 'female'}
> student.marks
< 94.4
> student.marks = "A";
< 'A'
> student.marks
< 'A'
> student
< {name: 'shradha', age: 23, marks: 'A', city: 'Mumbai', gender: 'female'}
>
```

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```
Elements Console Sources Network Performance >>
Filter Default levels No Issues
> student.city
< 'Mumbai'
> student
< {name: 'shradha', age: 23, marks: 94.4, city: 'Mumbai'}
> student.gender
< undefined
> student.gender = "female";
< 'female'
> student.gender
< 'female'
> student
< {name: 'shradha', age: 23, marks: 94.4, city: 'Mumbai', gender: 'female'}
> student.marks
< 94.4
> student.marks = "A";
< 'A'
> student.marks
< 'A'
> student
< {name: 'shradha', age: 23, marks: 'A', city: 'Mumbai', gender: 'female'}
> student.marks = [99, 89, 74];
< (3) [99, 89, 74]
> student
< {name: 'shradha', age: 23, marks: Array(3), city: 'Mumbai', gender: 'female'}
>
```

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Add/ Update Value

- Change the city to "Mumbai"
- Add a new property, gender: "Female"
- Change the marks to "A"

```
const student = {  
  name: "shradha",  
  age: 23,  
  marks: 94.4,  
  city: "Delhi"  
};
```

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delete obj.keyName ;



Object of Objects

Storing information of multiple students

```
const classInfo = {  
  aman : {  
    grade: "A+",  
    city: "Delhi"  
  },  
  shradha : {  
    grade: "A",  
    city: "Pune"  
  },  
  karan : {  
    grade: "O",  
    city: "Mumbai"  
  }  
};
```



Array of Objects

Storing information of multiple students

array
↓
objects

```
const classInfo = [  
  {  
    name: "aman",  
    grade: "A+",  
    city: "Delhi"  
  },  
  {  
    name: "shradha",  
    grade: "A+",  
    city: "Pune"  
  },  
  {  
    name: "karan",  
    grade: "O",  
    city: "Mumbai"  
  }  
];
```



<> index.html

JS app.js X

JS app.js > classInfo > city

```
14  // };
15
16  const classInfo = [
17    {
18      name: "aman",
19      grade: "A+",
20      city: "Delhi"
21    },
22    {
23      name: "shradha",
24      grade: "A",
25      city: "Pune"
26    },
27    {
28      name: "karan",
29      grade: "O",
30      city: "Mumbai"
31    }
32  ];
```



```
Elements Console Sources Network Performance >>
top Filter Default levels No Issues
> classInfo[0]
< ▶ {name: 'aman', grade: 'A+', city: 'Delhi'}
> classInfo[1]
< ▶ {name: 'shradha', grade: 'A', city: 'Pune'}
> classInfo[2]
< ▶ {name: 'karan', grade: 'O', city: 'Mumbai'}
> classInfo[1].name
< 'shradha'
> classInfo[1].grade
< 'A'
> classInfo[1].city
< 'Pune'
> classInfo[1].city = "Gurgaon";
< 'Gurgaon'
> classInfo[1]
< ▶ {name: 'shradha', grade: 'A', city: 'Gurgaon'}
> classInfo[1].gender = "female";
< 'female'
> classInfo[1]
< ▶ {name: 'shradha', grade: 'A', city: 'Gurgaon', gender: 'female'}
```

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Math Object

Properties

Math.PI

Math.E

Methods

Math.abs(n)

Math.pow(a, b)

Math.floor(n)

Math.ceil(n)

Math.random()

$\sin \theta$
 $\tan \theta$
 $\log$
 $\pi = 3.14$
 e



JS x +

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Elements Console Sources Network Performance >> | ⚙️ ⋮ ✕

top Filter Default levels No Issues ⚙️

```
> Math
< ▶ Math {abs: f, acos: f, acosh: f, asin: f, asinh: f, ...}
> Math.PI
< 3.141592653589793
> Math.E
< 2.718281828459045
> Math.abs(12);
< 12
> Math.abs(-12);
< 12
> Math.abs(-12.5);
< 12.5
> Math.pow(2, 4);
< 16
> Math.pow(2, 5);
< 32
> |
```



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Random Integers

From 1 to 10

Step1: `let num = Math.random();` 0.46747741318127045

Step2: `num = num * 10;`
4.674774131812704

Step3: `num = Math.floor(num);`
4

Step4: `num = num + 1;`
5

JS



Math.random() → 0 to 1 (1x)

Random Integers

From 1 to 10

Step1 : `let num = Math.random();` 0.46747741318127045

Step2 : `num = num * 10;`
4.674774131812704

Step3 : `num = Math.floor(num);`
4

Step4 : `num = num + 1;`
5



```
Elements Console Sources Network >>
top Filter Default levels No Issues
< 3.411372252248379
> let step3 = Math.floor(step2);
< undefined
> step3
< 3
> Math.random()
< 0.13170837266726543
> Math.random()
< 0.7181749051543687
> Math.random() * 10
< 0.9790236479152492
> Math.random() * 10
< 4.379401052554943
> Math.random() * 10
< 1.33346473416599
> Math.random() * 10
< 3.6254132489831736
> Math.random() * 10
< 6.599688334953573
> Math.floor( Math.random() * 10 );
< 7
> |
```

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Random Integers

From 1 to 10

```
let random = Math.floor(Math.random() * 10) + 1;  
undefined  
random  
4
```



1 to 10

7

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Guessing Game

User enters a max number & then tries to **guess a random generated number** between 1 to max.

```
user = prompt() // 10  
// random generate
```



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JS app.js

✕

JS app.js > ...

```
1  const max = prompt("enter the max number");
2
3  const random = Math.floor(Math.random() * max) + 1;
4
5  let guess = prompt("guess the number");
6
7  while(true) {
8      if(guess == "quit") {
9          console.log("user quit");
10         break;
11     }
12
13     if(guess == random) {
14         console.log("you are right! congrats!!");
15         break;
16     } else {
17         guess = prompt("your guess was wrong. please try again");
18     }
19 }
```



<> index.html

JS app.js

✕

JS app.js > ...

```
1  const max = prompt("enter the max number");
2
3  const random = Math.floor(Math.random() * max) + 1;
4
5  let guess = prompt("guess the number");
6
7  while(true) {
8      if(guess == "quit") {
9          console.log("user quit");
10         break;
11     }
12
13     if(guess == random) {
14         console.log("you are right! congrats!! random number was", random);
15         break;
16     } else if(guess < random) {
17         guess = prompt("your guess was too small. please try again");
18     } else {
19         guess = prompt("your guess was too large. please try again");
20     }
21 }
```



JS (Part 5)

Practice Questions

Qs1. Create a program that generates a random number representing a dice roll.
[The number should be between 1 and 6].

Qs2. Create an object representing a car that stores the following properties for the car: name, model, color.
Print the car's name.

Qs3. Create an object Person with their name, age and city.
Edit their city's original value to change it to "New York".
Add a new property country and set it to the United States.